

As shown in Table 1, each attribute/parameter has both a geodetic representation and a civic representation.

Table 1

Attribute	Geodetic	Civic
Position	59.3, 18.6	Stockholm
Elevation	8 metres	Third floor
Heading	142 degrees	To the city
Speed	12 km/h	Running
Orientation	55 degrees	North-West

Web storage

In HTML, to store user data on a local machine, we were using JavaScript cookies. To avoid that, HTML5 has introduced Web storage, with which websites themselves store user data on a local machine.

The advantages of Web storage, as compared to cookies, are:

- More secure
- Faster
- Stores a larger amount of data
- The stored data is not sent with every server request.

It is only included when asked for. This is a big advantage of HTML5 Web storage over cookies.

There are two types of Web storage objects:

- 1) Local – this stores data with no expiration date.
- 2) Session – this stores data for one session only.

How it works: The *localStorage* and *sessionStorage* objects create a *key = value* pair.

An example is: *key* = "Name", *value* = "Palak"

These are stored as strings but can be converted, if required, by using JavaScript functions like *parseFloat()* and *parseFloat()*.

Given below is the syntax for using Web storage objects:

Storing a Value:

```
• localStorage.setItem("key1", "value1");
• localStorage["key1"] = "value1";
```

Getting a Value:

```
• alert(localStorage.getItem("key1"));
• alert(localStorage["key1"]);
```

Remove a Value:

```
• removeItem("key1");
```

Remove All Values:

```
• localStorage.clear();
```

Application Cache (AppCache)

Using HTML5 Apache, we can make a Web application work offline without an Internet connection. All browsers, besides IE, can use *AppCache* (at this point in time).

The advantages of Application Cache are:

- Enables browsing Web pages offline
- Pages load faster

- Results in less load for servers

The *cache manifest* file is a simple text file that lists the resources the browser should cache for offline access. The *manifest* attribute can be included on the document's HTML tag, as follows:

```
<html manifest="test.appcache">
...
</html>
```

It should be on all the pages that you want to cache.

The application pages that are cached will remain unless:

1. The user clears them out.
2. The manifest has been modified.
3. The cache is updated.

Video

Until HTML5 was launched, there was no uniform standard for showing video on Web pages. Most of the videos were shown through different plugins like Flash. But HTML5 specifies a standard way to show the video on a Web page by using a video element.

Currently, it supports three video formats for the video element, as shown in Table 2.

Table 2

Format	IE	Firefox	Opera	Chrome	Safari
Ogg	No	3.5+	10.5+	5.0+	No
MPEG4	No	No	No	5.0+	3.0+
WebM	No	No	10.6+	6.0+	No

The example given below shows the use of this video element:

```
<!DOCTYPE HTML>
<html>
<body>
```

```
<video src=" video.ogg" width="320" height="240"
controls="controls">
```

This browser does not support the video element.

```
</video>

</body>
</html>
```

This example uses an Ogg file, and will work in Firefox, Opera and Chrome. To make the video work in Safari and future versions of Chrome, we must add an MPEG4 and WebM file.